

Project#6

OpenCL Linear Regression

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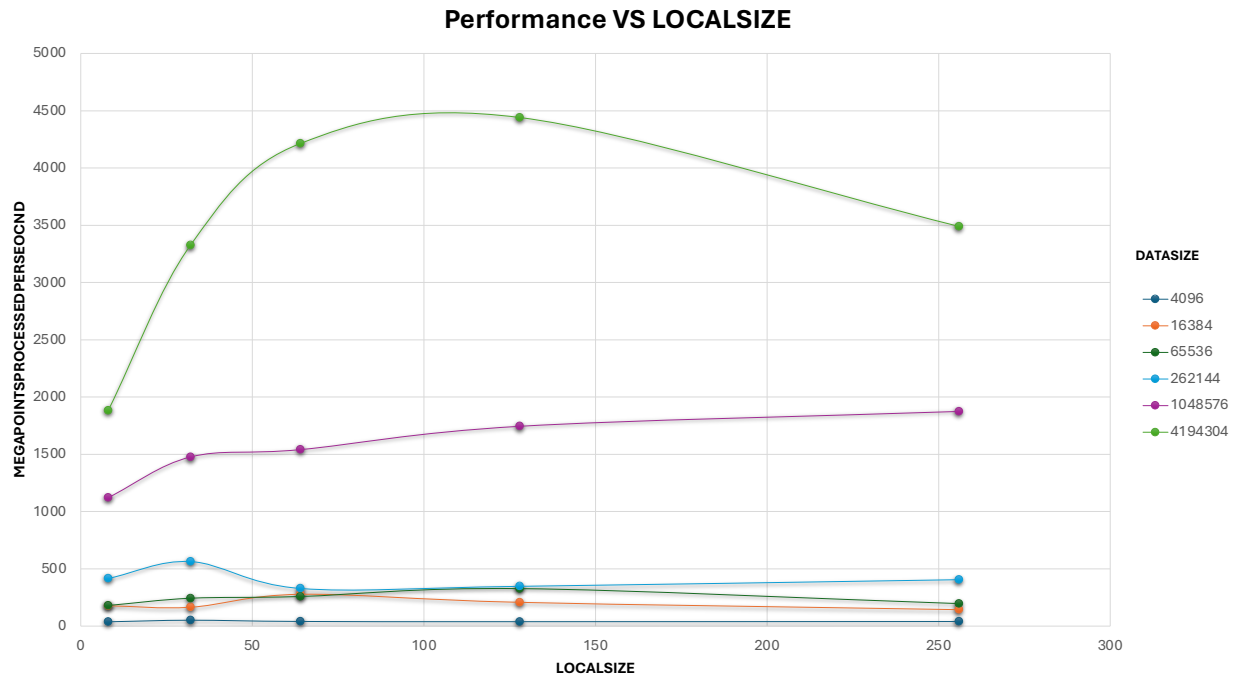
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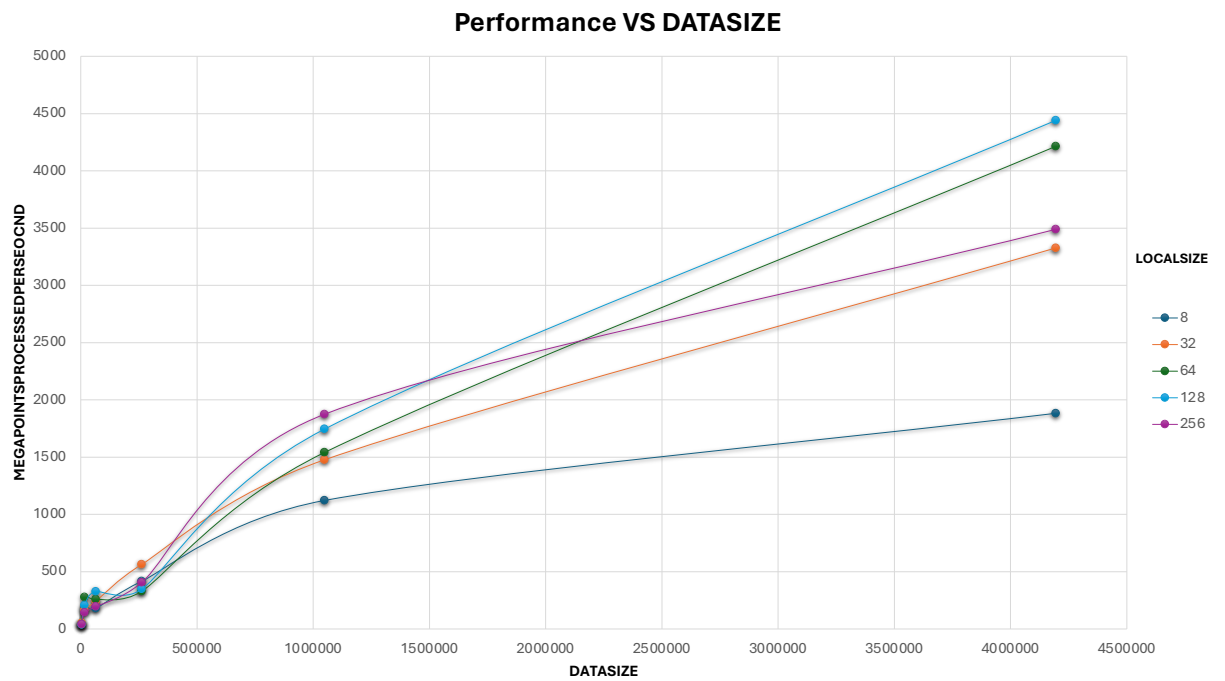
1. **What machine you ran this on.**
 - I ran code on rabbit server.
2. **M and B values**
 - $M = 5$
 - $B = 7$
3. **Show the table and graphs**
 - Table

m	b	ARRAYSIZE	LOCALSIZE	MegaPointsProcessedPerSecond
5	6.99	4096	8	37.74
5	6.99	4096	32	51.39
5	6.99	4096	64	40.68
5	6.99	4096	128	38.75
5	6.99	4096	256	40.18
5	7	16384	8	177.78
5	7	16384	32	164.91
5	7	16384	64	277.76
5	7	16384	128	207.6
5	7	16384	256	143.51
5	7	65536	8	180.2
5	7	65536	32	243.44
5	7	65536	64	258.65
5	7	65536	128	327.52
5	7	65536	256	196.41
5	7	262144	8	416.16
5	7	262144	32	563.43
5	7	262144	64	328.75
5	7	262144	128	347.3
5	7	262144	256	405.45
5	7	1048576	8	1122.09
5	7	1048576	32	1477.09
5	7	1048576	64	1540.95
5	7	1048576	128	1744.77
5	7	1048576	256	1874.44
5.01	7	4194304	8	1883.07
5.01	7	4194304	32	3326.77
5.01	7	4194304	64	4213.89
5.01	7	4194304	128	4440.47
5.01	7	4194304	256	3489.36

- Performance VS LOCALSIZE Graph



- Performance VS DATASIZE Graph



4. What patterns are you seeing in the performance curves? What difference does the size of data make? What difference does the size of each work-group make?

- **Performance VS LocalSize graph**

- The graph shows that performance improves as DataSize increases.
- For DataSize under 262144, the performance remains relatively low.
- When DataSize reaches 4194304, performance hits the maximum, but there is a noticeable curve if the LocalSize exceeds 128.

- **Performance vs. DataSize Graph**

- The graph illustrates how performance changes with different LocalSize settings.
- Each LocalSize graphs show linear growth as DataSize increases.
- LocalSize 128 shows the best performance, followed by 64 and 256.
- LocalSize 8 exhibits minimal performance improvement.

5. Why do you think the patterns look this way?

- **Performance vs. LocalSize Graph**

- Larger DataSize enhances performance dramatically.
 - Larger data sizes allow more extensive use of parallel processing capabilities in GPU.
 - With larger datasets, the relative overhead for managing individual tasks (like kernel launches and synchronization) is minimized.
 - Larger datasets involve more extensive computational tasks that take longer to process, reducing the frequency of context switching between tasks.
- Performance gains from increasing the LocalSize start to curve at a certain point.
 - This shows an optimal balance where hardware resources are fully utilized without excessive overhead.
 - Beyond this point, further increases in LocalSize do not show significant performance improvements and may lead to performance degradation due to increased synchronization costs and memory contention.

- **Performance vs. DataSize Graph**

- Performance is notably suboptimal when LocalSize is smaller than 32.
 - Since there are 32-threads each workgroup, it does not fully get advantage of parallel processing.
 - Not fully exploiting the available computational resources, leading to higher overheads and less efficient execution.
- Performance improves significantly as the LocalSize exceeds 32.
 - Reducing time spent on task management and memory access by using additional threads swapping in.
- LocalSize with 128 and 64 shows the better performance than 256
 - As I mentioned in analyzing Performance VS LocalSize graph, too many threads working on the same calculation may lead to performance degradation due to increase in synchronization costs and memory contention